

Having given the necessary introductions about complexity classes, let me dispel another myth about the power of quantum computers. People often incorrectly assume that a quantum computer will be able to solve all the problems in class NP in polynomial time. This is not what the experts believe.

Figure 1 shows a probable Venn diagram of some of the complexity classes, including the quantum complexity class BQP (Bounded-error Quantum Polynomial time). It is the class of problems that can be solved by a quantum computer in polynomial time with an error probability of at most 1/3. From Figure 1, we can have an idea about the kind of problems that can be solved efficiently by a quantum computer. It will definitely outperform any classical computer but, at the same time, the wide consensus among experts is that at least some of the problems in class NP will not be solvable by a quantum computer in polynomial time.

Now let us consider undecidable problems. These problems (could be thought of as unsolvable problems, though this is not a mathematically precise definition) are decision problems (the answer to such problems is either 'yes' or 'no') for which no algorithm exists to help us get an answer. An example is the Post correspondence problem. No classical computer can solve this problem algorithmically. What about quantum computers? Sadly, the problem remains unsolvable even if we use a quantum computer. So an important observation about quantum computers is the following. As far as 'computability' is considered, quantum computers do not have any advantage over classical computers. If a problem is undecidable (unsolvable) by a classical computer, it remains the same for quantum computers. At the same time, quantum computers will efficiently solve a lot of problems not feasible for classical computers.

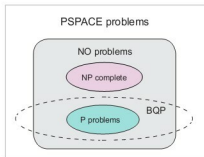


Figure 1: The complexity class BQP (Image credit: Wikipedia)

The progress made so far

Now that we know about the extra power that will be provided by fully functional quantum computers, let us discuss the different companies and research institutes that have made significant advances in quantum computing. As it can be a game changing technology, all the tech giants have invested heavily in quantum computing. Big players like Google, IBM, Microsoft, HP, Intel, Toshiba, etc, all have their own divisions to carry out research on quantum computing. Each one of these companies has contributed either to the hardware or the software for quantum computing.

In 2016, MIT (Massachusetts Institute of Technology) and the University of Innsbruck, Austria, together developed the first functional quantum computer. Though that quantum computer could only factorise relatively small numbers

like 15 using Shor's algorithm, it was a tremendous achievement as far as quantum computing was concerned. The next big milestone came in 2019. Google claimed quantum supremacy in October 2019. The tech giant developed a 53 qubit quantum processor called Sycamore, which it claimed achieved quantum supremacy by completing a computational task in 200 seconds. That computation would have taken at least 10,000 years for even the fastest classical supercomputer we have today, claimed Google. The report about the experiment, published in the famous journal *Nature*, can be found at <https://www.nature.com/articles/s41586-019-1666-5#Abs1>. Figure 2 shows Google's Sycamore quantum processor mounted on a cryostat.

Though the news of such a quantum processor made quite a sensation in the media, the claim of achieving quantum supremacy was questioned by many. IBM claimed that the same task could be completed in under 2.5 days in a classical computer. The company also claimed that further improvements might reduce the time taken for the solution even further. Details regarding the claims made by IBM can be found at <https://www.ibm.com/blogs/research/2019/10/on-quantum-supremacy/>. IBM has also touched significant milestones as far as quantum computing is concerned. It has already developed tools like Qiskit (an open source SDK

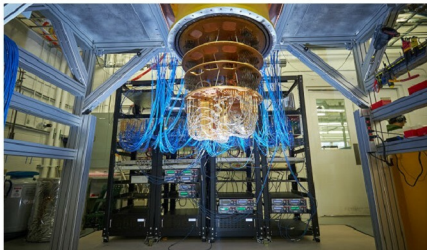


Figure 2: Google's Sycamore processor (Photo credit: Rocco Ceselin, photo obtained from the Google AI blog)